CARL ANDREI D. TORRES
CPE21S1
CPE-031 - Visualizations and Data Analysis

Instructions:

1. Watch the video:
2. The Power in Effective Data Storytelling – Malavica Sridhar | TEDxUIUC
3. Listen carefully to how the speaker explains the role of data, visuals, and storytelling.
4. Answer the guide questions in complete sentences. Be concise but thoughtful.
5. Submit your answers through a pdf file.

GUIDE QUESTIONS & ANSWERS:

1. **What is the "power" in data storytelling that the speaker emphasizes, and why does she believe it matters in influencing decisions or actions?**

   The speaker emphasizes that the true power of data storytelling lies in revealing a "plot twist" which is a compelling piece of data evidence that fundamentally changes a human behavior or business strategy itself. She points out that despite the massive amount of data produced globally, less than half a percent (0.5%) is ever actually utilized. Without a clear, focused "quest" or problem statement, the vast majority of the datasets remain untouched or lead to wasted analytical effort. An effective storytelling bridges this gap by turning raw facts into a meaning ful narrative that directly guides our decision-makers toward actionable solutions.

2. **How does she define or illustrate the role of audience awareness in crafting a data story?**

   In her analysis of the 1992 IBM Simon Personal Communicator, she shows that IBM failed to aware the audience by ignoring what the users actually wanted, such as affordability and long battery life. To address the decision-makers effectively strategize in the business, she describes how her team created a "ghost decks" by drawing imaginary graphs on paper before conducting an analysis. This practice helps her team to debate with the stakeholders whether

those mock visuals would genuinely answer the desired questions before investing hours in data storyboarding.

3. **What example(s) does she give of a visualization or story gone wrong and what are the root causes of the failure according to her?**

   The speaker highlights the multi-plot television shows as a prime example of the storyboarding gone wrong when they juggle ten competing storylines at once. She explains that the viewers give up by the second episode because the narrative lacks a single, clear mission storylines to follow. Similarly, the 1992 IBM Simon Personal Communicator as a product failure because it crammed in the features like the touchscreen tech without solving a clear user need itself. She stated that the problem that they didn't establish "what they are solving for" which waste significant resources going down unproductive or confusing paths.

4. **Which "preattentive" or design principles does she highlight as important for making charts communicate more clearly (for example: color, size, position, emphasis)?**

   She presented her three-part narrative framework which are the Problem, the Solution, and the Inputs. The first step is that it requires precisely defining the core problem or statement before looking at the dataset needed. The second step focuses on the mapping of the protagonist's game plan with tools like the ghost decks to ensure the output is actionable and feasible. Finally, the third step ensures that the underlying raw data inputs are complete, accurate, and reliable enough to support our analysis.

5. **Based on her advice, how would you redesign one of your own visualizations (or a chart you've seen) to better tell a story? What would you change and why?**

   Applying her ideas to redesign one of my visualizations. First, I would find a "Quest" that I would need to solve and present the dataset given on the chart. Next, I would create like a "Ghost Deck" to test whether the proposed structure directly supports our "Quest" before loading our dataset. Finally, I would eliminate the outliers or distracting secondary metrics to focus on the entire visual on highlighting a single, unexpected "Plot Twist" like the speaker said in the video.



Please have the honor of pledge in the comment section or at the bottom of your activity document/exam or any form of file submission.

I affirm that I will not give or receive any unauthorized help on this activity/exam and that all work will be my own.